Advanced Research Grant (ARG) - Use-Inspired Basic Research

Technical Document (Single PDF Upload)

Upload ONE consolidated PDF covering all sections below. Font size 12 (Times New Roman or Arial), single-spaced; Maximum 20 pages (including figures/tables/bibliography);

Note to the PI: The Full Proposal should remain within the objectives, scope and research direction outlined in the shortlisted pre-proposal. The Full Proposal should provide greater detail, refinement and methodological elaboration; Substantial changes in the objectives, scope, research questions or overall direction of the proposed research are not permitted. The Full Proposal should clearly demonstrate continuity with, and further development of, the shortlisted pre-proposal.

1. Use-Inspired Context, Scientific Rationale & Knowledge Gap

Define the real-world or use-inspired context motivating the research and the specific challenge or opportunity it presents. Explain the underlying scientific rationale, current state-of-the-art (India and abroad, with proper references), and the fundamental knowledge gap that must be addressed.

2. Research Objectives, Hypotheses & Success Metrics

List the specific research objectives and hypotheses, each clearly traceable to the knowledge gap and use-inspired challenge identified in Section 1. Define measurable scientific and, where appropriate, utility-oriented success metrics or target values against relevant baselines.

3. Scientific Novelty, Advancement of Knowledge & Utility

You may explain the originality of the proposed research and how it will advance fundamental understanding while addressing a clearly defined use-inspired need. Describe the limitations of current approaches and the anticipated improvement in the relevant technological baseline, such as efficiency, cost, performance, sensitivity, reliability, sustainability, or other appropriate parameters. Where preliminary data or proof-of-concept exists, summarize it here.

4. Research Approach, Methodology, Integration & Feasibility

You may describe the research strategy and high-level execution logic, broken into phases or work packages. Where relevant, explain benchmarking against existing approaches and how performance, cost, efficiency or other utility parameters will be assessed.

5. Expected Knowledge Outcomes, Utility, Impact & Future Translation

- Advancement of fundamental knowledge and resolution of the identified knowledge gap
- Expected improvement over the relevant scientific/technological baseline and potential utility
- Potential longer-term technological, societal or economic benefits and future translational pathways

6. Timeline & Milestones

- Provide a year-wise Bar (Gantt) chart of activities and milestones, aligned with the proposed objectives

7. Team, Expertise & Institutional Facilities

7.1 Expertise available with the PI/Co-PI(s) relevant to executing this project.

7.2 Roles and responsibilities of all Investigators (summary table).

Investigator	Role	Key Responsibilities

7.3 Five Key publications (with impact factor) by the Investigators relevant to this proposal's theme, in the last 5 years.

7.4 Infrastructural facilities and equipment available at the host institute(s)/department(s) relevant to this project.

8. Justification for proposed budget: (Provide a clear and concise justification for the proposed expenditure under each major budget head, for all the investigators)

- Equipment (if any): (Specify the equipment proposed, its purpose and technical relevance to the project. Explain why the equipment is essential for the proposed research and why existing institutional facilities or shared resources cannot adequately meet the requirement. Indicate, where relevant, how the equipment will be utilized during the project)
- Recurring head (including Research Personnel): Justify the requirement under each relevant recurring head, including research personnel, consumables, travel and contingency.

9. Current Financial Support (Other Than This Proposal)

Project Title	Funding Agency	Amount	Duration	Overlap with This Proposal (if any)

10. Name and address of experts / institution interested in the subject/outcome of the project

S. No.	Name of Expert	Affiliation	Email	Area of Expertise

11. Bibliography